PRESS RELEASE

People who eat a late dinner may gain weight

Washington, DC | June 11, 2020

Eating a late dinner may contribute to weight gain and high blood sugar, according to a small study published in the Endocrine Society's *Journal of Clinical Endocrinology & Metabolism.*

Over 2.1 billion adults are estimated to have overweight or obesity which make health complications like diabetes and high blood pressure more likely. Some studies suggest that consuming calories later in the day is associated with obesity and metabolic syndrome.

"This study sheds new light on how eating a late dinner worsens glucose tolerance and reduces the amount of fat burned. The effect of late eating varies greatly between people and depends on their usual bedtime," said the study's corresponding author Jonathan C. Jun, M.D., of the Johns Hopkins University School of Medicine in Baltimore, M.d. "This shows that some people might be more vulnerable to late eating than others. If the metabolic effects we observed with a single meal keep occurring chronically, then late eating could lead to consequences such as diabetes or obesity."

The researchers studied 20 healthy volunteers (10 men and 10 women) to see how they metabolized dinner eaten at 10 p.m. compared to 6 p.m. The volunteers all went to bed at 11 p.m. The researchers found that blood sugar levels were higher, and the amount of ingested fat burned was lower with the later dinner, even when the same meal was provided at the two different times.

“On average, the peak glucose level after late dinner was about 18 percent higher, and the amount of fat burned overnight decreased by about 10 percent compared to eating an earlier dinner. The effects we have seen in healthy volunteers might be more pronounced in people with obesity or diabetes, who already have a compromised metabolism,” said the study’s first author Chenjuan Gu, M.D., Ph.D., of the Johns Hopkins University.

This is not the first study to show effects of late eating, but it is one of the most detailed. Participants wore activity trackers, had blood sampling every hour while staying in a lab, underwent sleep studies and body fat scans, and ate food that contained non-radioactive labels so that the rate of fat burning (oxidation) could be determined.

“We still need to do more experiments to see if these effects continue over time, and if they are caused more by behavior (such as sleeping soon after a meal) or by the body’s circadian rhythms,” Jun said.

Other authors include Nga Brereton, Amy Schweitzer, Daisy Duan, Luu V. Pham and Vsevolod Y. Polotsky of the Johns Hopkins University; and Matthew Cotter, Elisabet Børsheim and Robert R. Wolfe of the University of Arkansas for Medical Sciences in Little Rock, Ark.

The study was supported by the Foundation for the National Institutes of Health and the American Heart Association.

The manuscript, *“Metabolic Effects of Late Dinner in Healthy Volunteers – A Randomized Crossover Clinical Trial,”* was published online, ahead of print.

About the Endocrine Society

Endocrinologists are at the core of solving the most pressing health problems of our time, including diabetes, obesity, infertility, bone health, and hormone-related cancers. The Endocrine Society is the largest global organization of scientists devoted to hormone research and physicians who care for people with hormone-related conditions.

With more than 18,000 members in 133 countries, the Society serves as the voice of the endocrine field. Through its renowned journals and ENDO, the world's largest endocrine meeting, the Society accelerates hormone research, advances clinical excellence in endocrinology, and advocates for evidence-based policies on behalf of the global endocrine community. To learn more, visit our **online newsroom**.

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